

DARPA-SN-25-57
Crystal Substrate Bonding Technologies and Algorithms (CRYSTAL) Webinar
April 2, 2025

The Defense Advanced Research Projects Agency (DARPA) Defense Sciences Office (DSO) is sponsoring a live virtual webinar to provide information about the Crystal Substrate Bonding Technologies and Algorithms (CRYSTAL) Advanced Research Concepts (ARC) Opportunity (DARPA-EA-25-02-04). The webinar will include information specific to the CRYSTAL ARC Opportunity as well as an overview of the ARC initiative. Registrants will hear from Government personnel and leading researchers in related fields and participate in question-and-answer sessions. The virtual webinar will be held on Wednesday, April 2, 2025. Registration is required for all attendees of the webinar.

To maximize the pool of innovative research concepts, DARPA strongly encourages participation in these events and subsequent solicitations by non-traditional performers, including small businesses, academic and research institutions, and first-time Government contractors.

(CRYSTAL) OPPORTUNITY DESCRIPTION

Overview: DARPA/DSO has issued an ARC Opportunity, inviting submissions of abstracts for innovative exploratory research concepts in the technical domain of wafer bonding of thin-film crystals (<https://www.darpa.mil/ARC>). ARC Opportunities are designed to allow an individual researcher the opportunity and time to focus on nascent, paradigm-shifting ideas for national security. This ARC Opportunity, titled CRYSTAL, is issued under the master ARC Exploration Announcement (EA), DARPA-EA-25-02.

Background: Bonded single crystal thin film multi-functional materials (electro-optic, acousto-electric, acousto-optic, magneto-optic or multi-ferroic materials) are vital for diverse sensing and communications technologies (integrated quantum, photonic, terahertz, radio frequency, and actuator platforms). Wafer bonding onto compatible substrates is the critical step for integrating single crystal thin films into multi-functional devices and systems. There is currently no method to analytically investigate wafer bonding processes. Brute force experimentation is required for initial process development, and subsequent changes in materials or device parameters necessitate significant additional trial and error, incurring huge costs, and prolonging timeframes, thus limiting exploration of novel material systems. The ability to predictively model wafer bonding of thin film crystals would rapidly accelerate the research and development of multi-functional materials, and their fabrication and scalable integration in diverse applications.

Solicitation Question: To accelerate development and integration of multi-functional materials, how do we create generalizable models to explore thin film crystal bonding onto suitable substrates under diverse real-world process conditions and parameters?

Scope: The CRYSTAL ARC Opportunity seeks proposals that explore modeling/simulation frameworks to fundamentally understand how process conditions affect crystal-substrate wafer bond formation and interface properties between thin film multi-functional materials and

compatible substrates. This ARC Opportunity asks proposers to create a bonding model for a material-substrate platform of the proposer's choice, which exhibits multi-functional properties.

REGISTRATION INFORMATION

The virtual webinar will be held on Wednesday, April 2, 2025, from 11:00 a.m. to 3:00 p.m. ET with virtual check-in beginning at 10:30 a.m. ET.

Participants must register to attend the webinar through the registration website: <https://events.sa-meetings.com/CRYSTALWebinar/>. There is no fee for the webinar.

Registration Opens: As of publication of this announcement.

Registration Closes: Tuesday, April 1, 2025, at 12:00 p.m. ET, or when capacity is reached, whichever comes first.

If interested individuals cannot register because registration is closed or capacity has been reached, instructions for being added to the waitlist are provided on the website. If slots remain open after registration closes or become available due to cancellations, the slots will be filled on a first come, first served basis from the waitlist. Individuals requesting to attend in excess of the capacity limitation will also be added to the waitlist.

Attendance is open to individuals who are U.S. citizens, U.S. permanent residents, and foreign nationals. All registrants who are not U.S. citizens must complete and submit either a DARPA Form 60 ("U.S. Permanent Resident and Foreign National Visit Request" – e.g., industry or academia) or an Official Visit Request (foreign government personnel only) through their country's embassy based in Washington, DC, no later than 5:00 p.m. ET on March 26, 2025. Form 60 submission instructions are provided on the registration website and in the registration confirmation email. Contact your embassy staff for assistance in submitting the Official Visit Request.

DISCLAIMERS

Attendance at the CRYSTAL webinar is voluntary and is not required to propose to DARPA-EA-25-02-04. The webinar does not constitute a formal solicitation for proposals or abstracts. This announcement is issued solely for information and program planning purposes and is not a Request for Information (RFI). Since this is not an RFI, the Government will not accept submissions against this notice. DARPA will not provide reimbursement for costs incurred to participate in this webinar.

CONTACT INFORMATION

Registration Website: <https://events.sa-meetings.com/CRYSTALWebinar/>

DARPA ARC Website: <https://www.darpa.mil/ARC>

POC: CRYSTAL Administrative Team at CRYSTAL@darpa.mil

Administrative questions should be submitted to CRYSTAL@darpa.mil. Please include “CRYSTAL Webinar” in the subject line in all correspondence. Inquiries that do not include this subject line may not receive a response.

For those new to DARPA or national security, DARPA makes available a free, comprehensive resource via DARPAConnect on how to do business with the agency. In addition to DARPA 101 materials, relevant preparatory modules include “Making the Most of a Proposers Day” and “Understanding DARPA Broad Agency Announcements.” Registration and access are free at www.darpaconnect.us.